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SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

Detection of Cyber Attack in Network using Machine Learning Techniques

INT 500 Internship

Final Review

Team Members

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Project Guide

1. Ms.k Yamini

Introduction

- Sophisticated cyber attacks demand stronger network security solutions.
- Traditional methods like firewalls and encryption have limitations against evolving threats.
- Machine learning offers adaptability for improved intrusion detection.
- Machine learning allows knowledge bases to stay updated, leading to better detection rates.
- The rise of network applications and the Internet of Things (IoT) necessitates robust security.
- Newer datasets like CICIDS2017 are needed to address limitations of older ones like KDD99.

Research/Product survey

Title

- Detection of Cyber Attack in Network using Machine Learning Techniques

Journal

- IEEE

Algorithm

- SVM

Purpose

- Identify and prevent cyber attacks in network traffic

Dataset

- CIC-IDS2017

Advantages

- Adaptable, efficient, improved accuracy.

Disadvantages

- Data quality, false positives, explainability.

Parameters

- Labeled data, pre-processing, training, evaluation, deployment.

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Title

- Machine Learning for Application-Layer Attack Detection

Journal

- IEEE

Algorithm

- Graph-based segmentation and dynamic programming (for future work)

Purpose

- Identify and prevent application-layer cyber attacks

Dataset

- Network traffic captured from user activity

Advantages

- Early attack detection, adaptability to new threats

Disadvantages

- Data quality dependence, potential false positives

Parameters

- Resource usage (CPU, memory, disk), user activity, network traffic metrics

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Title

- Predicting Network Attack Patterns in SDN using Machine Learning Approach

Journal

- IEEE

Algorithm

- BayesNet (highest accuracy among tested algorithms)

Purpose

- Predict vulnerable hosts in SDN networks for security rule definition

Dataset

- Network attack data (various datasets used)

Advantages

- High prediction accuracy, improves network security

Disadvantages

- Dataset variance affects accuracy, requires historical data

Parameters

- Threshold (α), training/testing split ratio

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Title

- Intrusion Detection using Machine Learning Algorithm

Journal

- IEEE

Algorithm

- Support Vector Machine (SVM) for classification of cyber attacks.

Purpose

- To detect cyber attacks early and mitigate potential damage.

Dataset

- CICIDS2017, a modern dataset that includes information about new attack types.

Advantages

- Early detection of attacks, adaptability to new attack patterns, and automatic notification to security engineers.

Disadvantages

- Dependence on the quality and relevance of the training dataset, potential for false positives and negatives

Parameters

- Parameters of the SVM algorithm, features from the CICIDS2017 dataset used for training and testing

Problem statement/Objectives

- Despite technological advancements, evolving cyber threats pose a continuous challenge to data security and network stability.
- Existing intrusion detection methods, like SVMs, show limitations in accurately identifying specific attack types, necessitating exploration of new and potentially more effective approaches.

DataSet

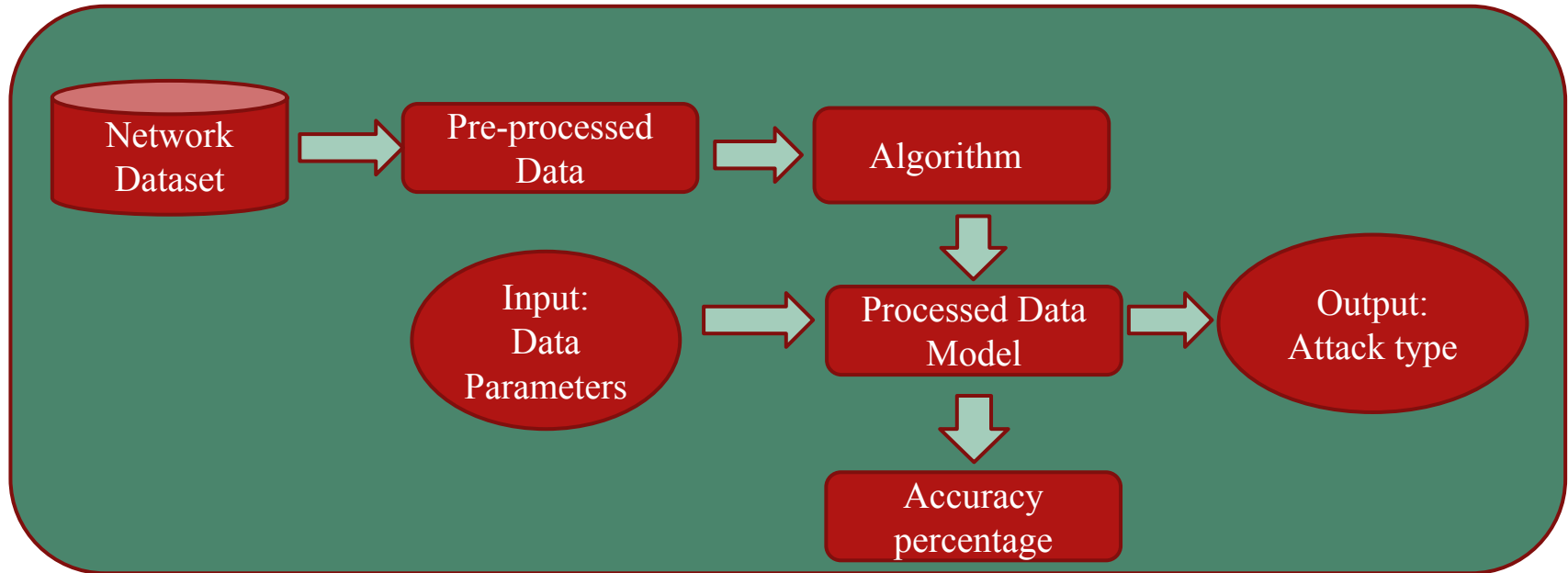
Dataset source :

<https://www.kaggle.com/code/shahtiham/unsw-nb15/notebook#Logistic-Regressoin-after-removing-outlie>

0	dur	175341	non-null	float64
1	proto	175341	non-null	int32
2	service	175341	non-null	int32
3	state	175341	non-null	int32
4	spkts	175341	non-null	int64
5	dpkts	175341	non-null	int64
6	sbytes	175341	non-null	int64
7	dbytes	175341	non-null	int64
8	rate	175341	non-null	float64
9	sload	175341	non-null	float64
10	dload	175341	non-null	float64
11	sloss	175341	non-null	int64
12	dloss	175341	non-null	int64
13	sinpkt	175341	non-null	float64
14	dinpkt	175341	non-null	float64
15	sjit	175341	non-null	float64
16	djit	175341	non-null	float64
17	swin	175341	non-null	int64
18	stcpb	175341	non-null	int64
19	dtcpb	175341	non-null	int64

20	dwin	175341	non-null	int64
21	tcprtt	175341	non-null	float64
22	synack	175341	non-null	float64
23	ackdat	175341	non-null	float64
24	smean	175341	non-null	int64
25	dmean	175341	non-null	int64
26	trans_depth	175341	non-null	int64
27	response_body_len	175341	non-null	int64
28	ct_src_dport_ltm	175341	non-null	int64
29	ct_dst_sport_ltm	175341	non-null	int64
30	is_ftp_login	175341	non-null	int64
31	ct_ftp_cmd	175341	non-null	int64
32	ct_flw_http_mthd	175341	non-null	int64
33	is_sm_ips_ports	175341	non-null	int64
34	attack_cat	175341	non-null	int32

Workflow/Methodology



Preprocessing Techniques

Data Encoding

- Data encoding transforms data into a different format for easier storage, processing, or transmission.
- In machine learning, it specifically focuses on converting categorical data (like qualities or categories) into numerical format.
- This is necessary because many machine learning algorithms only work with numerical inputs, and encoding allows you to use categorical data with them.

Outliers

- Identify outliers in datasets using visual inspection (plotting) or statistical techniques (Z-score, IQR).
- Handle outliers by removing them, transforming them, or using robust statistics/algorithms less sensitive to outliers.

Machine Learning Models

Decision Tree

A decision tree is one of the most powerful tools of supervised learning algorithms used for both classification and regression tasks. It builds a flowchart-like tree structure where each internal node denotes a test on an attribute, each branch represents an outcome of the test, and each leaf node (terminal node) holds a class label. It is constructed by recursively splitting the training data into subsets based on the values of the attributes until a stopping criterion is met, such as the maximum depth of the tree or the minimum number of samples required to split a node.

Logistic Regression

Logistic Regression is a statistical model used in machine learning for binary classification problems. It uses the logistic function to model a binary dependent variable. In simple terms, it predicts the probability of occurrence of an event by fitting data to a logistic function. Hence, it is also known as logit regression. Since it predicts the probability, its output values lie between 0 and 1 (as expected).

K-Nearest Neighbor(KNN) Algorithm

The K-Nearest Neighbors (KNN) algorithm is a type of instance-based learning method used for both classification and regression problems. The key idea behind KNN is that similar data points should have similar output labels.

Given a new, unknown instance, KNN goes through the entire dataset to find the k instances that are closest to the new instance. These k instances are the “nearest neighbors” of the new instance.

Modules

Software Requirements:

- **Python:** Install Python (preferably Python 3.x).
- **Scikit-learn (sklearn):** A powerful machine learning library.
- Jupyter Notebook or any other IDE.
- Pandas, NumPy, and Matplotlib for data manipulation and visualization.

Hardware Requirements:

- **Processor:** Multi-core CPU.
- **Memory (RAM):** At least 8GB.
- **Storage:** Adequate space for datasets and code.
- **Optional:** Graphics card (GPU) for faster deep learning training.
- **Network Interface Card (NIC)** for live network packet capture

Expected Outcome

Logistic Regression after removing outlier

```
In [43]: from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.model_selection import cross_val_score
import numpy as np
logr = LogisticRegression()
logr_cross = fit_and_evaluate(logr, xx_train , xx_test , yy_train , yy_test)

print('Logistic Regression Performance on the test set: Cross Validation Score = %.4f' % logr_cross)
```

Logistic Regression Performance on the test set: Cross Validation Score = 0.7147

```
In [44]: from sklearn.metrics import accuracy_score
y_pred = logr.predict(xl_test)
print("Accuracy on test dataset: ", accuracy_score(y1_test, y_pred))
```

Accuracy on test dataset: 0.5526037707866002

KNN:

```
In [46]: from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier(n_neighbors = 5, metric = 'minkowski', p = 2)
knn_cross = fit_and_evaluate(knn, xx_train , xx_test , yy_train , yy_test)

print('KNN Performance on the validation set: Cross Validation Score = %0.4f' % knn_cross)
```

KNN Performance on the validation set: Cross Validation Score = 0.7645

```
In [47]: yl_pred = knn.predict(xl_test)
print("Accuracy on test dataset: ", accuracy_score(yl_test, yl_pred))
```

Accuracy on test dataset: 0.41249389765360917

Decision Tree

```
in [50]: from sklearn.tree import DecisionTreeClassifier
         from sklearn.metrics import accuracy_score
         import matplotlib.pyplot as plt

         # Assuming dt is your Decision Tree model and it's already fitted with xx_train and yy_train

         # Predict on the training data
         y_train_pred = dt.predict(xx_train)

         # Evaluate accuracy on the training set
         try:
             train_accuracy = accuracy_score(yy_train, y_train_pred)
             print("Accuracy on train dataset: ", train_accuracy*100)
         except Exception as e:
             print("Error occurred while computing training accuracy:", e)
```

Accuracy on train dataset: 91.97545414545226

```
in [51]: from sklearn.metrics import accuracy_score

         # Evaluate accuracy on the test set
         try:
             accuracy = accuracy_score(yy_test, y_pred)
             print("Accuracy on test dataset: ", accuracy)
         except Exception as e:
             print("Error occurred while computing accuracy:", e)
```

Accuracy on test dataset: 0.8239730978346271

Streamlit

Home Page x FOURTH YEA x Untitled pre x Sample Proje x Untitled doc x Gemini x main - Stream x national inter x decision tree x +

localhost:8501

Deploy

Detection of Cyber Attack in Network using Machine Learning Techniques

enter a column number

You have selected: `DecisionTreeClassifier(criterion='entropy', random_state=42)`

Integer

37.00 - +

Detect

	dur	proto	service	state	spkts	dpkts	sbytes	dbytes	rate	load	dload
37	0.7837	113	0	2	10	8	564	354	21.6925	5,185.7837	3,164.553

Normal

value

6

ENG 13:31

References

- [1]Uyyala P. DETECTION OF CYBER ATTACK IN NETWORK USING MACHINE LEARNING TECHNIQUES. Journal of interdisciplinary cycle research. 2022;14(3)
- [2]Reddy D, Sajjan BT, Anusha M, Sadiq SJ, Shambulingappa HS. Detection of Cyber Attack in a Network using Machine Learning Techniques. International Journal of Advanced Scientific Innovation. 2021 May 27
- https://www.youtube.com/watch?v=hWbntQyMrI0&ab_channel=PankajVarmaDendukuri

**Thank
you**